

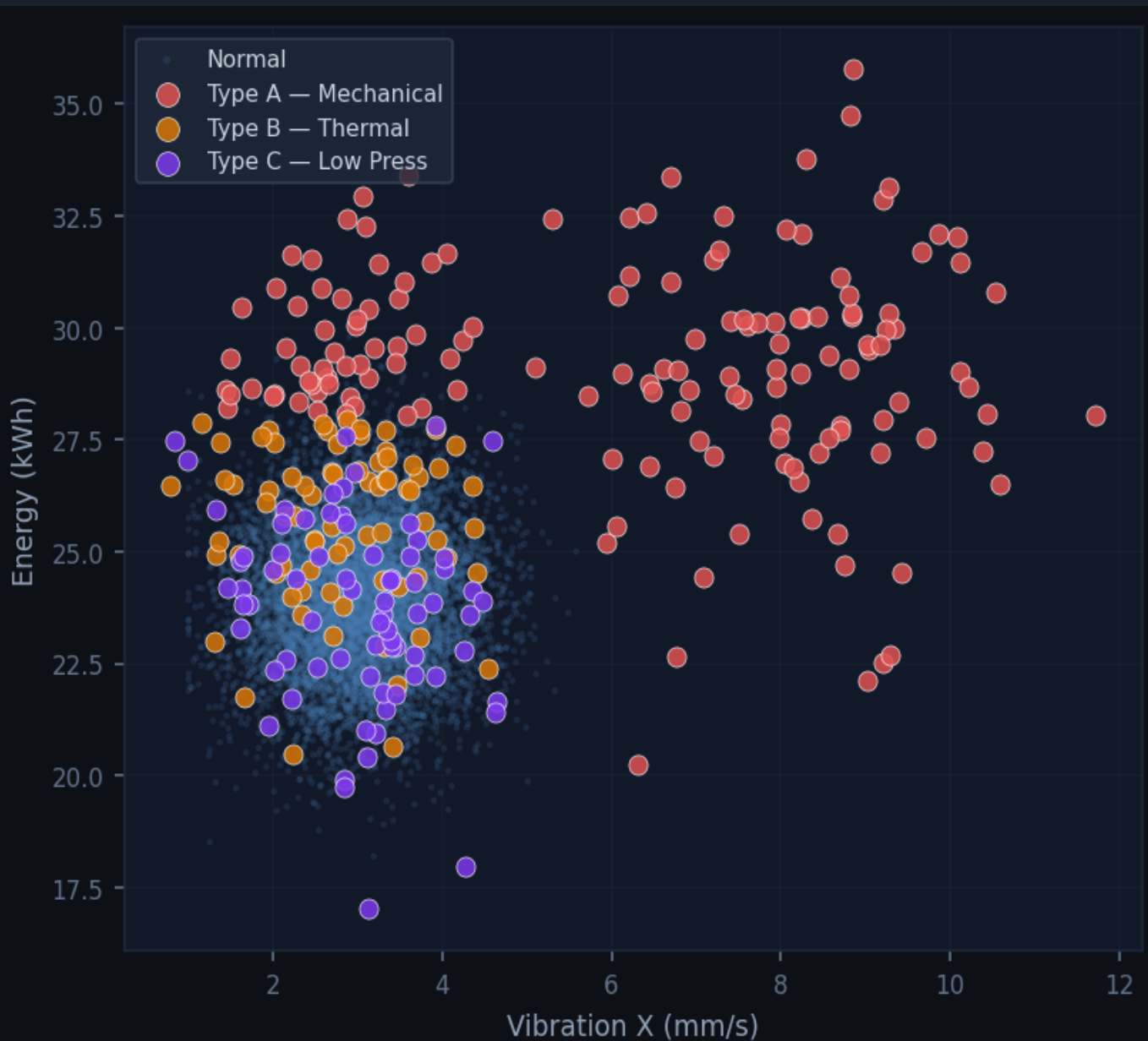
Press Anomaly Intelligence

Isolation Forest · Hot Stamping Line
Multi-type unsupervised detection

ISOLATION FOREST

UNSUPERVISED

SCIKIT-LEARN



10,312

Stamping Cycles

300

Anomalies

84.7%

Recall Score

3 Types

Detected

A boron steel blank at 950°C, pressed at 220+ tons, cooled under active lubrication — this process doesn't generate one type of failure.

TYPE A

Mechanical Failure

- Bearing wear on press axis
- Vibration X/Y spikes above 5.5 mm/s
- Energy draw increases with friction
- Detectable: high vibration + energy signature

TYPE B

Thermal Failure

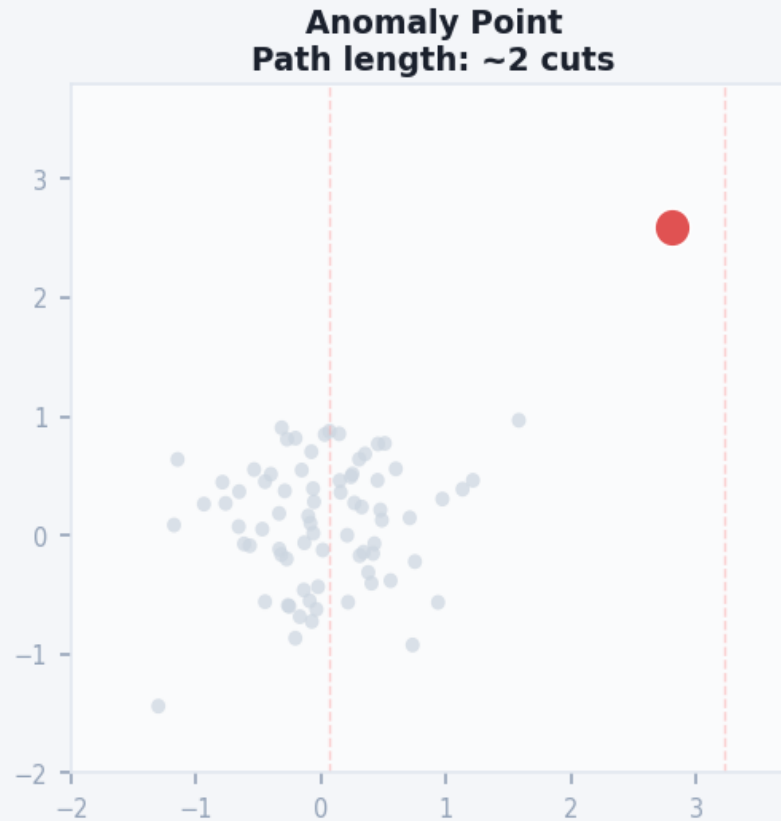
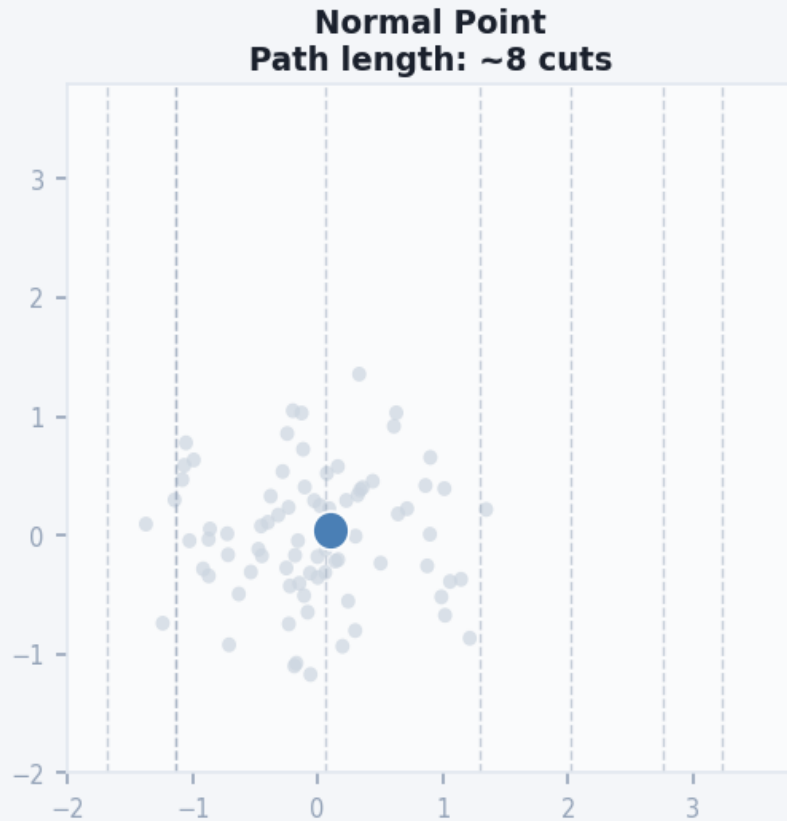
- Furnace fault or cooling system trip
- Blank arrives below forming temperature
- Extended cycle time as robot compensates
- Detectable: temp + cycle time co-elevation

TYPE C

Press Failure

- Die alignment drift or tonnage valve fault
- Press force drops to 178 ton vs 230 nominal
- Contact force follows — 128 kN vs 185 normal
- Hardest to detect: only 5% drop from normal mean

If I randomly slice this data space — how many cuts does it take to isolate this point?



-1.0 →

Shortest path

Highly anomalous

-0.53 →

Decision threshold

3rd percentile

0.0 →

Longest path

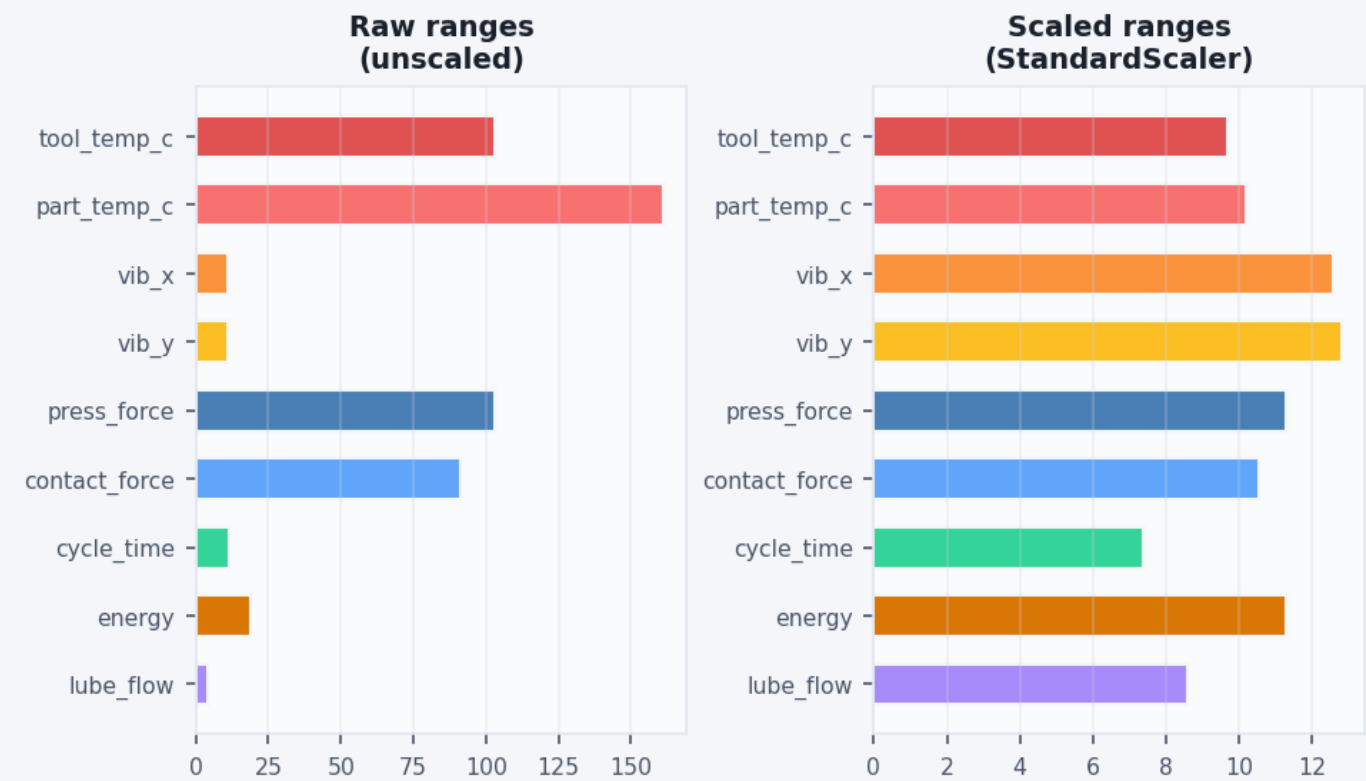
Highly normal

300 trees · Average path length = Anomaly Score

Short path = easy to isolate = anomalous. Long path = surrounded by neighbors = normal.

Nine sensors · Hot stamping press · 10,312 production cycles

Pyrometer	tool_temp_c	Normal: 420–440	Unit: °C	B	type driver
Pyrometer	part_temp_c	Normal: 820–880	Unit: °C	B	type driver
SCADA	vibration_x_mm_s	Normal: 1.5–5.0	Unit: mm/s	A	type driver
SCADA	vibration_y_mm_s	Normal: 1.5–5.0	Unit: mm/s	A	type driver
PLC	press_force_ton	Normal: 210–250	Unit: ton	C	type driver
PLC	contact_force_kn	Normal: 165–200	Unit: kN	C	type driver
MES	cycle_time_s	Normal: 15–22	Unit: s	B	type driver
MES	energy_kwh	Normal: 20–30	Unit: kWh	A	type driver
PLC	lubricant_flow_lmin	Normal: 3.5–6.5	Unit: L/min	—	type driver



Without scaling, tool_temp_c (100°C range) dominates random splits over lubricant_flow (4.4 L/min range). Every sensor must contribute equally.

The scaler is fitted on the training set and reused on all incoming cycles — refitting on new data resets the baseline.

Model Configuration

n_estimators

300

Stable average path lengths

contamination

0.03

Matches known 3% anomaly rate

bootstrap

True

Subsampling with replacement

random_state

42

Reproducibility

Score Interpretation: score_samples() → [-1, 0]



-1.0
Anomaly

-0.53
Threshold

0.0
Normal

Precision

0.8516

85.2% of alarms are real

Recall

0.8544

85.4% of anomalies caught

F1-Score

0.8530

Balanced across 3 types

Accuracy

0.9912

10,221 of 10,312 correct

Confusion Matrix (N = 10,312 cycles)

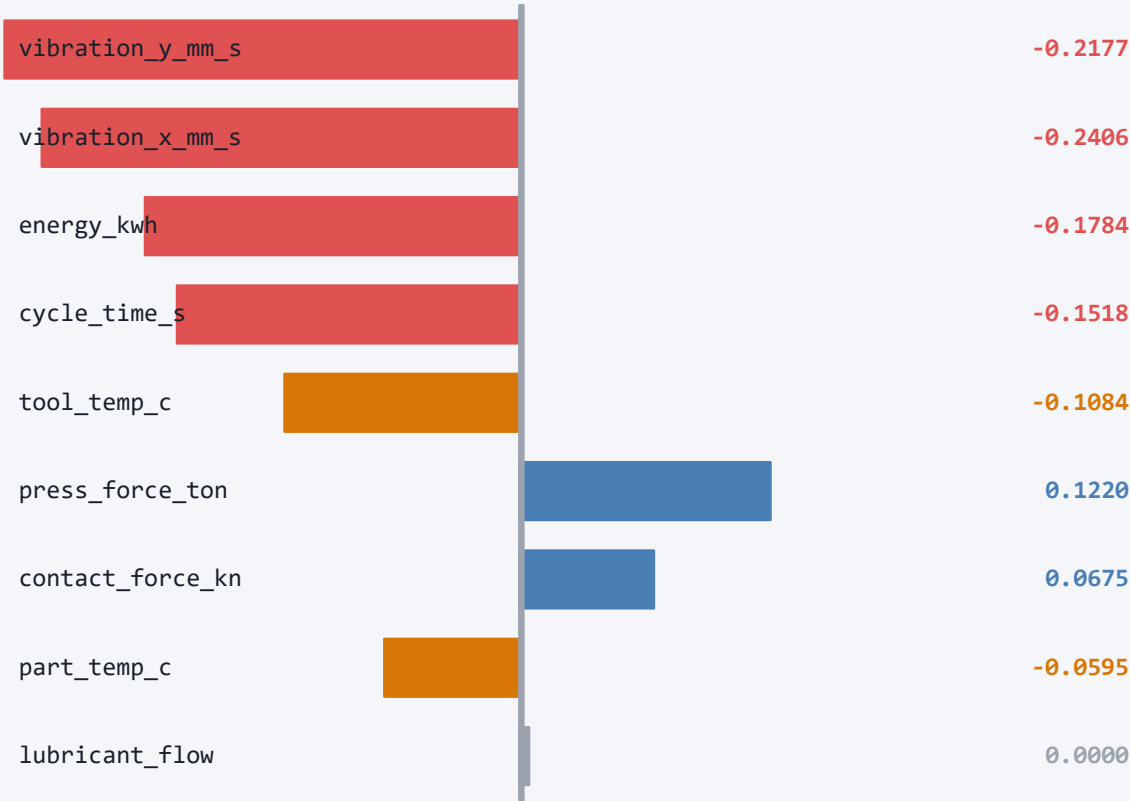
	Pred Normal	Pred Anomaly
Actual Normal	9,957 TN	46 FP
Actual Anomaly	45 FN	264 TP

Fix for 45 false negatives:

Add domain-rule post-filter: `press_force_ton < 200` → flag as Type C anomaly.
No retraining required — the model handles types A and B; the rule handles C.

The 45 false negatives are exclusively Type C — 5% force drop sits within the normal distribution overlap zone.

Feature → Anomaly Score Correlation



The IF score tells you severity. The sensor that drove the score tells you where to look.

TYPE A — Mechanical

Key sensors: `vibration_x_mm_s` · `vibration_y_mm_s` · `energy_kwh`

Profile: Vibration +50% · Energy +11% · Cycle time +13.5%



Stop press. Inspect spindle bearings and press head. Check tool run-out and axis guide lubrication. Measure vibration spectrum before restart.

TYPE B — Thermal

Key sensors: `tool_temp_c` · `part_temp_c` · `cycle_time_s`

Profile: Tool temp +2.2% · Part temp +1.3% · Cycle time +13.5%



Check furnace setpoint and pyrometer calibration. Verify blank heating time. Inspect cooling circuit flow rate. Review last 20 cycles for thermal drift trend.

TYPE C — Low-Pressure (Hardest to detect)

Key sensors: `press_force_ton` · `contact_force_kn`

Profile: Press force -5.0% (178 vs 230 ton) · Contact force -4.1%



Check tonnage valve calibration and die clamping sequence. Inspect die alignment on all four corners. Add post-filter rule: `press_force_ton < 200` → auto-flag.

A — Stable

tool_temp_c	430 °C
part_temp_c	850 °C
vibration_x	3.0 mm/s
vibration_y	3.2 mm/s
press_force	230 ton
contact_force	185 kN
cycle_time	18.0 s

✓ NORMAL



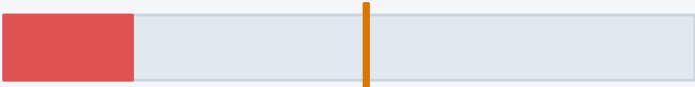
IF Score: -0.3625

← threshold -0.5326

B — Mechanical

tool_temp_c	435 °C
part_temp_c	855 °C
vibration_x	10.5 mm/s
vibration_y	9.8 mm/s
press_force	232 ton
contact_force	188 kN
cycle_time	23.0 s

🚨 CRITICAL



IF Score: -0.6735

← threshold -0.5326

C — Low Press

tool_temp_c	428 °C
part_temp_c	848 °C
vibration_x	3.1 mm/s
vibration_y	3.0 mm/s
press_force	178 ton
contact_force	128 kN
cycle_time	21.0 s

⚠️ ANOMALY



IF Score: -0.5701

← threshold -0.5326

Threshold = -0.5326 (3rd pct). Score below threshold → anomaly. Farther from 0 → higher severity.

300 trees. Three failure modes. One decision.

The elegant core of Isolation Forest is philosophical as much as mathematical: instead of defining what normal looks like, it measures how easy it is to separate a point from all others.

Real industrial anomaly detection is rarely a solved problem. It is a conversation between the data, the domain, and the algorithm.

Download the notebook · Vary contamination from 0.01 to 0.07 · See how F1 changes.

Add the `press_force_ton` domain rule · Measure the recall gain.

Because the best model is the one that knows its own limits.

